

State and input observability of multi-agent systems: A necessary and sufficient condition

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ABSTRACT

This paper considers the strong observability of multi-agent systems where a subset of agents are modified by unknown inputs. Leveraging a minimum separator decomposition, a necessary and sufficient condition for the state and input observability of multi-agent systems is established from a graph-theoretic perspective. The condition is formed by some special subgraphs and can be checked without complex computation. Besides, when the condition is not satisfied, it is shown that the state and input observability can be recovered by adding additional sensors to a subset of agents. A feasible sensor addition procedure is provided. Numerical examples corroborate the practical merits of the theoretical results.

1. Introduction

Multi-agent systems have been extensively studied, where agents exchange local information with their neighbors and reach global consensus in finite time. One fundamental problem is observability analysis of the system. If a multi-agent system is observable, by collecting local output measurements, the initial states of the whole system can be reconstructed [1,2]. The problem can be solved by traditional observability theory of linear systems, including PBH criterion or rank test of observability matrix. Since the structure of a multi-agent system can be uniquely represented by the graph, an alternative approach is formulating sufficient and necessary conditions from a graph-theoretic perspective [3,4]. Specifically, graph partitions, dominate sets and maximum matching can be utilized to determine whether a given multi-agent system is observable.

With the application of multi-agent systems in open and distributed scenario, the security issues may endanger the normal operation of the system [5]. Agents in the system might be hijacked by attackers and influenced by unknown inputs. For a multi-agent system with unknown interference, the initial states can be recovered if the system is strongly observable [6]. This indeed, boils down to a state and input observability problem.

The state and input observability of the system is closely related to fault detection and isolation, resilient control, and state estimation. Given a state and input observable system, observer can be designed in the presence unknown inputs, and both states and unknown inputs can

be reconstructed from the output of the system. In [7], unknown input observers are designed for linear parameter varying (LPV) systems, and the known rank conditions are relaxed. Similar to observability analysis, algebraic and graph-theoretic approaches can be utilized to characterizing state and input observability. In order to recover unknown inputs, the zero-input response and zero-state response should be decoupled from each other, and rank test of observability matrix is insufficient for state and input observability. In [8], a necessary and sufficient condition is formulated by observability matrix and invertibility matrix, and the result was extended to time-varying network systems. The strong observability problem can be also solved by utilizing Rosenbrock system matrix [9]. If the column normal-rank of Rosenbrock matrix is full, and there exists no invariant zeros, the corresponding system is state and input observable. Alternatively, the conditions for state and input observability were also obtained from a graph-theoretic perspective. In [10], a necessary and sufficient condition was proposed, and the condition can be checked by finding particular subgraphs. The results was extended to partial state and input observability [11], and any given part of states can be recovered in the presence of unknown inputs.

A multi-agent system can be uniquely represented by a graph, therefore, the state and input observability of multi-agent system can also be analyzed through the graph. If the graph topology of a multi-agent system is sufficiently connected and the number of attacked agents is upper bounded, the system is observable in the presence of unknown

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inputs [12]. In [13], a similar condition was derived from a system-theoretic perspective. It is shown the graph connectivity can counteract the influence of unknown inputs. Specifically, letting the number of attacked agents be upper bounded by f , if the vertex connectivity of the system is $f + 1$, the state of the system can be reconstructed from local outputs. The conditions in [12,13] are expressed in terms of the number of attacked agents and the vertex connectivity. However, it remains unclear whether and how the number of agents which can be directly measured influence the observability. Furthermore, if a multi-agent system is unobservable, how to recover the state and input observability is not well studied. Aiming to solve these issues, the state and input observability problem of multi-agent system is revisited in the paper. Leveraging a minimum separator decomposition, a necessary and sufficient condition for the state and input observability in terms of measurable agents is provided. If the condition is not met, additional sensors can be implemented to recover the observability. The effect of extra measurable agents on the state and input observability is investigated. Moreover, to improve the performance of systems in a malicious environment, two-hop information is utilized. Adjacent agents with high security level can share their observation, which increases the number of directly measured agents. The contributions are summarized as follows:

- A necessary and sufficient condition for state and input observability is provided, and it establishes the relationship between the observability and the graph topology of multi-agent systems.
- It is shown that observation sharing among adjacent agents can improve state and input observability without additional hardware implementation.
- Given a multi-agent system which is not state and input observable, by adding extra sensors to agents, the observability can be recovered. The location and the number of extra sensors is specified, and two numerical examples illustrate the effectiveness of the proposed method.

The rest of this paper is organized as follows. Section 2 briefly introduces graph theory preliminaries, definitions of relevant concepts, and known results used in the paper. Section 3 formulates the problem considered in the paper. Section 4 presents our main results. A necessary and sufficient condition for state and input observability are provided, and it can be utilized to recover the observability of sparsely connected graphs. Conclusions are drawn in Section 5.

2. Notation and related results

2.1. Notation

In this paper, matrices (column vectors) are denoted by upper-(lower-) case boldface letters. Sets are denoted using calligraphic letter. The cardinality of set $\mathcal{X}$ is denoted by $|\mathcal{X}|$. The symbol $e_{i,n}$ represents the $n \times 1$ vector with a 1 in its i th position and 0's elsewhere. The symbol I_N represents the $N \times N$ identity matrix; $\mathbf{0}$ represents zero matrices of appropriate dimensions.

A graph is denoted by $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$ is a set of vertices and $\mathcal{E}$ is a set of ordered pairs of different vertices, called edges. The vertices in the set $\mathcal{N}_i = \{v_j | (v_j, v_i) \in \mathcal{E}\}$ are called neighbors of vertex v_i . A path is a sequence of edges without duplicated vertices. If the start vertex and the end vertex of a path are connected, the graph is called a circle. If two paths share no common vertices, they are vertex-disjoint. A path is a $\mathcal{Y}$ -topped path if its end vertex belongs to $\mathcal{Y}$. A path P is a $\mathcal{V}_1 - \mathcal{V}_2$ path if its start vertex belongs to set $\mathcal{V}_1$ and its end vertex belongs to set $\mathcal{V}_2$. If the only vertices of P belonging to $\mathcal{V}_1 \cup \mathcal{V}_2$ are its start and its end vertices, the path is a direct $\mathcal{V}_1 - \mathcal{V}_2$ path. A set of r vertex-disjoint $\mathcal{V}_1 - \mathcal{V}_2$ paths are called an r -linking from $\mathcal{V}_1$ to $\mathcal{V}_2$. The maximal number of vertex-disjoint $\mathcal{V}_1 - \mathcal{V}_2$ paths is denoted by $\rho(\mathcal{V}_1, \mathcal{V}_2)$, and a $\rho(\mathcal{V}_1, \mathcal{V}_2)$ -linking is called maximum $\mathcal{V}_1 - \mathcal{V}_2$ linking. If the removal of a subset $S \subseteq \mathcal{V}$ disconnects every path from

$\mathcal{V}_1$ to $\mathcal{V}_2$, the subset S is a separator between $\mathcal{V}_1$ to $\mathcal{V}_2$. A separator of minimal cardinality is a minimum separator, and its cardinality is $\rho(\mathcal{V}_1, \mathcal{V}_2)$. All minimum separators between $\mathcal{V}_1$ to $\mathcal{V}_2$ form the essential vertices, denoted by $\mathcal{V}_{ess}(\mathcal{V}_1, \mathcal{V}_2)$. A special minimum separator S^* is defined as the end vertices of all direct $\mathcal{V}_1 - \mathcal{V}_{ess}(\mathcal{V}_1, \mathcal{V}_2)$ paths.

2.2. System modeling and preliminaries

Consider a multi-agent system consists of n agents, which adopts a linear iteration rule for coordination. The underlying communication graph $\mathcal{G}$ formed by the agents is assumed connected and fixed. It is further assumed that every agent can observe its own state as well as the states of the neighbors. When the system operates normally (i.e., in the absence of attacks), each agent updates its state using a weighted combination of observed states. At each time step k , agent i performs the following update

$$x_i[k+1] = w_{ii}x_i[k] + \sum_{j \in \mathcal{N}_i} w_{ij}x_j[k], \quad \forall i = 1, 2, \dots, n \quad (1)$$

where $x_i \in \mathbb{R}$ is the state of agent i , and w_{ij} is the weight attached to the influence agent j has on agent i .

Concatenating the states of all the agents in a column vector, the update strategy of the entire system can be described as

$$\mathbf{x}[k+1] = \underbrace{\begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1} & w_{n2} & \cdots & w_{nn} \end{bmatrix}}_W \mathbf{x}[k], \quad k \in \mathbb{N} \quad (2)$$

where $w_{ij} = 0$ if $j \notin \mathcal{N}_i \cup i$, and W is the weight matrix.

Suppose a number of agents in the system are hijacked by attackers, whose indices are collected in $\mathcal{I}$ with $|\mathcal{I}| = p$. The remaining agents operate normally and are represented by $\bar{\mathcal{I}} = \mathcal{V} \setminus \mathcal{I}$. The agents in set $\bar{\mathcal{I}}$ update their states according to (1). Nevertheless, for every agent in set $\mathcal{I}$, the state is manipulated by attackers in the following fashion

$$x_i[k+1] = w_{ii}x_i[k] + \sum_{x_j \in \mathcal{N}_i} w_{ij}x_j[k] + u_i, \quad \forall i \in \mathcal{I} \quad (3)$$

where u_i denotes the unknown datum injected by attackers.

Let $\mathcal{I} = \{i_1, i_2, \dots, i_p\}$, and the multi-agent system under attacks can be compactly described by

$$\mathbf{x}[k+1] = W\mathbf{x}[k] + \underbrace{[e_{i_1,n}, e_{i_2,n}, \dots, e_{i_p,n}]}_B \underbrace{\begin{bmatrix} u_{i_1}[k] \\ u_{i_2}[k] \\ \vdots \\ u_{i_p}[k] \end{bmatrix}}_{\mathbf{u}[k]} \quad (4)$$

where $B \in \mathbb{R}^{n \times p}$ is the unknown input matrix, and $\mathbf{u}[k] \in \mathbb{R}^{p \times 1}$ is the unknown input vector.

To model the value measured by agent i , let $\mathcal{O}_i = \{o_j | o_j \in \mathcal{N}_i \cup i\}$, and $|\mathcal{O}_i| = m$. The states of agents in $\mathcal{O}_i$ can be directly measured

$$\mathbf{y}_i[k] = \underbrace{\begin{bmatrix} e_{o_1,n}^T \\ e_{o_2,n}^T \\ \vdots \\ e_{o_m,n}^T \end{bmatrix}}_{C_i} \mathbf{x}[k], \quad \forall i = 1, 2, \dots, n \quad (5)$$

where $\mathbf{y}_i \in \mathbb{R}^{m \times 1}$ is the states observed by agent i , and $C_i \in \mathbb{R}^{m \times n}$ consists of basis vectors denoting agent i and its neighbors.

Combining (4) with (5), the state-space representation of agent i is given as follows

$$\mathbf{x}[k+1] = W\mathbf{x}[k] + B\mathbf{u}[k] \quad (6a)$$

$$y_i[k] = C_i x[k] \quad (6b)$$

Denoting the output of the system (6) over n time-steps by

$$\begin{bmatrix} y_i[0] \\ y_i[1] \\ y_i[2] \\ \vdots \\ y_i[n-1] \end{bmatrix} = \underbrace{\begin{bmatrix} C_i \\ C_i W \\ C_i W^2 \\ \vdots \\ C_i W^{n-1} \end{bmatrix}}_{\text{observability matrix}} x[0] + \underbrace{\begin{bmatrix} 0 & 0 & \cdots & 0 \\ C_i B & 0 & \cdots & 0 \\ C_i W B & C_i B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ C_i W^{n-2} B & C_i W^{n-3} B & \cdots & C_i B \end{bmatrix}}_{\text{invertibility matrix}} \begin{bmatrix} u[0] \\ u[1] \\ u[2] \\ \vdots \\ u[n-2] \end{bmatrix} \quad (7)$$

If the columns of observability matrix and invertibility matrix are linear independent, the initial states $x[0]$ of the multi-agent system and unknown inputs $\{u[k], k \in \{0, 1, \dots, n-2\}\}$ can be uniquely computed from the sequence $\{y_i[k], k \in \{0, 1, \dots, n-1\}\}$.

Definition 1 (State and Input Observability). Consider the multi-agent system with unknown input in (6). For agent i , the system is state and input observable (SIO) if $y_i[k] = 0$ for all k implies $x[0] = 0$, and $u[k] = 0$.

2.3. Related results

For a linear system with state-space representation (A, B, C, D) , the matrix pencil is defined as follows

$$P(z) = \begin{bmatrix} A - zI_n & B \\ C & D \end{bmatrix} \quad (8)$$

where $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{m \times n}$, $D \in \mathbb{R}^{m \times p}$, $I_n \in \mathbb{R}^{n \times n}$, and $z \in \mathbb{C}$. The normal-rank of $P(z)$ is defined as $\max_{z \in \mathbb{C}}(\text{rank}(P(z)))$.

Definition 2 (Invariant Zero). The complex number $z_0 \in \mathbb{C}$ is an invariant zero of the system with matrix pencil $P(z)$ in (8) if $\text{rank}(P(z_0))$ is less than the normal-rank of $P(z)$.

Lemma 1 ([10]). The system with matrix pencil $P(z)$ in (8) is SIO if and only if $\forall z \in \mathbb{C}$, $\text{rank}(P(z)) = n + p$.

To analyze the system from a graph-theoretic perspective, we can use techniques from control theory pertaining to linear structured systems [10]. Specifically, if the non-zero entries of the system $(A^\lambda, B^\lambda, C^\lambda, D^\lambda)$ are free parameters, where λ denotes all the non-zero parameters, then the system is structured.

Since the entry of structured system $(A^\lambda, B^\lambda, C^\lambda, D^\lambda)$ is either zero or nonzero, the system can be uniquely represented by graph $\mathcal{G}_A(\mathcal{V}, \mathcal{E})$. The vertex set $\mathcal{V}$ contains the input vertex set $\mathcal{U} = \{u_1, u_2, \dots, u_p\}$, the state vertex set $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$, and the output vertex set $\mathcal{Y} = \{y_1, y_2, \dots, y_m\}$. Hence, set $\mathcal{V}$ consists of $n + p + m$ vertices. The edge set $\mathcal{E}$ of the graph is described by $\mathcal{E}_A \cup \mathcal{E}_B \cup \mathcal{E}_C \cup \mathcal{E}_D$ with $\mathcal{E}_A = \{(x_j, x_i) | [A^\lambda]_{ij} \neq 0\}$, $\mathcal{E}_B = \{(u_j, x_i) | [B^\lambda]_{ij} \neq 0\}$, $\mathcal{E}_C = \{(x_j, y_i) | [C^\lambda]_{ij} \neq 0\}$, and $\mathcal{E}_D = \{(u_j, y_i) | [D^\lambda]_{ij} \neq 0\}$.

Example 1. For the system defined by the following matrices, according to the definition, the associated digraph $\mathcal{G}_A(\mathcal{V}, \mathcal{E})$ can be constructed. The topology structure is illustrated in Fig. 1.

$$A^\lambda = \begin{bmatrix} \lambda_1 & \lambda_2 \\ \lambda_3 & 0 \end{bmatrix}, \quad B^\lambda = \begin{bmatrix} \lambda_4 \\ 0 \end{bmatrix},$$

$$C^\lambda = [\lambda_5 \quad 0], \quad D^\lambda = [\lambda_6].$$

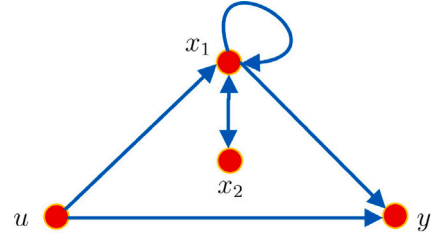


Fig. 1. Digraph representing a structured system.

Given a structured system $(A^\lambda, B^\lambda, C^\lambda, D^\lambda)$ with a random choice of free parameter λ^* . If the system has t invariant zeros, then the structured system with almost any choice of λ still has t invariant zeros (except for some situations with Lebesgue zero) [14]. How to compute the number of invariant zeros from the graph of the structured system was given as follows.

Lemma 2 ([15]). The normal-rank of $P(z)$ is equal to n plus the maximum size of a linking from $\mathcal{U}$ to $\mathcal{Y}$.

Lemma 3 ([15]). If matrix pencil $P(z)$ is invertible, the number of invariant zeros is generically equal to $n + p$ minus the minimum number of edges in a maximum size linking from $\mathcal{U}$ to $\mathcal{Y}$.

Lemma 4 ([15]). If matrix pencil $P(z)$ has full column rank, even after the deletion of an arbitrary row, then the number of invariant zeros is equal to n minus the maximum number of vertices in the disjoint union of a linking of size p from $\mathcal{U}$ to $\mathcal{Y}$, a $\mathcal{Y}$ -topped path, and a set of circles in $\mathcal{X}$.

3. Problem formulation

The states of agents in set $\mathcal{O}_i$ can be directly measured by agent i , and the remaining agents are denoted by $\bar{\mathcal{O}}_i = \mathcal{V} \setminus \mathcal{O}_i$. Now set $\mathcal{V}$ can be divided into four mutually disjoint subsets, i.e., $\mathcal{I}\mathcal{O}_i$, $\bar{\mathcal{I}}\mathcal{O}_i$, $\mathcal{I}\bar{\mathcal{O}}_i$, and $\bar{\mathcal{I}}\bar{\mathcal{O}}_i$. To facilitate analysis, $x[k]$ is the state vector of all agents at time-step k , and assume without loss of generality that $x[k]$ is of the form $x[k] = [x_{\bar{\mathcal{I}}\bar{\mathcal{O}}_i}^T[k] \ x_{\mathcal{I}\bar{\mathcal{O}}_i}^T[k] \ x_{\mathcal{I}\mathcal{O}_i}^T[k] \ x_{\bar{\mathcal{I}}\mathcal{O}_i}^T[k]]^T$. The state-space representation of the system in (6) becomes the following form

$$\begin{aligned} x[k+1] &= \begin{bmatrix} W_{11} & W_{12} & W_{13} & W_{14} \\ W_{21} & W_{22} & W_{23} & W_{24} \\ W_{31} & W_{32} & W_{33} & W_{34} \\ W_{41} & W_{42} & W_{43} & W_{44} \end{bmatrix} x[k] + \begin{bmatrix} 0 & 0 \\ I_{|\bar{\mathcal{I}}\bar{\mathcal{O}}_i|} & 0 \\ 0 & 0 \\ 0 & I_{|\mathcal{I}\mathcal{O}_i|} \end{bmatrix} u[k], \\ y_i[k] &= \begin{bmatrix} 0 & 0 & I_{|\bar{\mathcal{I}}\mathcal{O}_i|} & 0 \\ 0 & 0 & 0 & I_{|\mathcal{I}\mathcal{O}_i|} \end{bmatrix} x[k] \end{aligned} \quad (9)$$

where W_{ij} represents the sub-matrix block, $u[k]$ is an error vector injected by attackers, and $y_i[k]$ is the output vector which can be directly measured by agent i .

The matrix pencil $P_i(z)$ of (9) is

$$P_i(z) = \begin{bmatrix} W - zI_n & B \\ C_i & 0 \end{bmatrix}. \quad (10)$$

From [12], it is known the invariant zeros of $P_i(z)$ are the same as the invariant zeros of the following matrix pencil

$$P_{i0}(z) = \begin{bmatrix} W_{11} - zI_{|\bar{\mathcal{I}}\bar{\mathcal{O}}_i|} & W_{12} \\ W_{31} & W_{32} \end{bmatrix}. \quad (11)$$

Remark 1. Every nonzero entry of B and C_i is equal to one, rather than free parameters. Therefore, the original state-space representation $(W, B, C_i, 0)$ is not a structured system. However, the entries of the weight matrix are free parameters, and the state-space representation $(W_{11}, W_{12}, W_{31}, W_{32})$ is structured.

According to [12,13], the strong observability of multi-agent system is determined by vertex connectivity and the number of attacked agents. However, the relationship between observability and the directly measurable agents remains unclear. The aim of this paper is to give graphical conditions to determine whether a multi-agent system is generically SIO. These conditions can be further utilized to recover the state and input observability by including additional measurable vertices.

4. Main results

4.1. Conditions of state and input observability

The network of the multi-agent system is represented by graph $\mathcal{G}$: the vertex set represents all agents in the system, the edge set represents whether agents have influence on each other. Since matrix pencil $P_0(z)$ has the same invariant zeros with $P_i(z)$, the agents in $\mathcal{I}\mathcal{O}_i$ do not influence the number of invariant zeros. The agents in sets $\bar{\mathcal{I}}\mathcal{O}_i$, $\mathcal{I}\mathcal{O}_i$, and $\bar{\mathcal{I}}\mathcal{O}_i$ are state vertices, input vertices, and output vertices, respectively. The graph $\mathcal{G}_0$ which represents the structured system $(\mathbf{W}_{11}, \mathbf{W}_{12}, \mathbf{W}_{31}, \mathbf{W}_{32})$ can be constructed from $\mathcal{G}$ as follows: remove all the agents in set $\mathcal{I}\mathcal{O}_i$ as well as the edges connected with them, remove all the incoming edges of vertices in $\bar{\mathcal{I}}\mathcal{O}_i$, and the outgoing edges of vertices in $\bar{\mathcal{I}}\mathcal{O}_i$, and add a self-loop to every vertex in $\bar{\mathcal{I}}\mathcal{O}_i$.

The construction of graph $\mathcal{G}_0$ includes removal of nodes and edges in graph $\mathcal{G}$, and could lead to disconnection of the graph or isolation of vertex. To make our discussion rigorous, some standing assumptions on graph $\mathcal{G}_0$ are made as follows:

Assumption 1. Every vertex in $\bar{\mathcal{I}}\mathcal{O}_i$ is the begin vertex of an edge that ends in a vertex in $\bar{\mathcal{I}}\mathcal{O}_i$; Every vertex in $\bar{\mathcal{I}}\mathcal{O}_i$ is the end vertex of an edge that starts in $\bar{\mathcal{I}}\mathcal{O}_i$; Every vertex in $\bar{\mathcal{I}}\mathcal{O}_i$ is contained in a path from $\mathcal{I}\mathcal{O}_i$ to $\bar{\mathcal{I}}\mathcal{O}_i$.

If the matrix pencil has full column rank, even after deleting an arbitrary row, the number of invariant zeros can be computed from the graph according to Lemma 4. A sufficient condition of SIO is provided.

Proposition 1. Given the multi-agent system (9) with unknown inputs, which has the same number of invariant zeros with the structured linear system $(\mathbf{W}_{11}, \mathbf{W}_{12}, \mathbf{W}_{31}, \mathbf{W}_{32})$, the system is generically SIO if in the associated digraph $\mathcal{G}_0$ the following two conditions hold

$$\rho(\mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i \setminus x_j) = |\mathcal{I}\mathcal{O}_i|, \quad \forall x_j \in \bar{\mathcal{I}}\mathcal{O}_i, \quad (12a)$$

$$\rho(x_j \cup \mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i) = |\mathcal{I}\mathcal{O}_i| + 1, \quad \forall x_j \in \bar{\mathcal{I}}\mathcal{O}_i. \quad (12b)$$

Proof. Now that $\rho(\mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i \setminus x_j) = |\mathcal{I}\mathcal{O}_i|$, $\forall x_j \in \bar{\mathcal{I}}\mathcal{O}_i$, and the maximal number of vertex-disjoint $\mathcal{I}\mathcal{O}_i - \bar{\mathcal{I}}\mathcal{O}_i \setminus x_j$ paths is $|\mathcal{I}\mathcal{O}_i|$. The number of vertex-disjoint paths is upper bounded by the number of vertices in set $\mathcal{I}\mathcal{O}_i$ and $\bar{\mathcal{I}}\mathcal{O}_i \setminus x_j$. Therefore, adding a vertex into set $\bar{\mathcal{I}}\mathcal{O}_i \setminus x_j$ will not increase the number of vertex-disjoint paths. it is clear that

$$\rho(\mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i) = |\mathcal{I}\mathcal{O}_i| \quad (13)$$

The normal rank of the matrix pencil is $|\bar{\mathcal{I}}\mathcal{O}_i| + |\mathcal{I}\mathcal{O}_i|$, therefore, the matrix pencil has full column rank.

After the deletion of an arbitrary row from $P_0(z)$, it is necessary to verify the matrix pencil still has full column rank. If a row corresponding to a state vertex is deleted, the vertex can be viewed as an input vertex. The number of the state vertex is $|\bar{\mathcal{I}}\mathcal{O}_i| - 1$, and $\rho(x_j \cup \mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i) = |\mathcal{I}\mathcal{O}_i| + 1$, $\forall x_j \in \bar{\mathcal{I}}\mathcal{O}_i$, thus the normal rank is still $|\bar{\mathcal{I}}\mathcal{O}_i| + |\mathcal{I}\mathcal{O}_i|$; If a row corresponding to an output matrix is deleted, the vertex is isolated from the graph. Due to $\rho(\mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i \setminus x_j) = |\mathcal{I}\mathcal{O}_i|$, $\forall x_j \in \bar{\mathcal{I}}\mathcal{O}_i$, the normal rank of the matrix pencil is $|\bar{\mathcal{I}}\mathcal{O}_i| + |\mathcal{I}\mathcal{O}_i|$.

Since every state vertex in the graph contains a self-loop, all vertices are included in the disjoint union. According to Lemma 4, the system has no invariant zeros. The system is SIO. $\square$

From Lemma 1, the system is generically SIO if and only if the corresponding matrix pencil has full column rank for all $z \in \mathbb{C}$. Given a structured system Σ_A , the corresponding graph is $\mathcal{G}_A$. The system is SIO if and only if $\rho(\mathcal{U}, \mathcal{Y}) = p$, and the system has no invariant zeros.

Remark 2. The size of a maximum linking from $\mathcal{U}$ to $\mathcal{Y}$ is $\rho(\mathcal{U}, \mathcal{Y})$. From Lemma 2, the normal-rank of the matrix pencil is equal to n plus the maximum size of a linking from $\mathcal{U}$ to $\mathcal{Y}$. Therefore, the normal rank is $n + \rho(\mathcal{U}, \mathcal{Y})$. If $\rho(\mathcal{U}, \mathcal{Y}) = p$, and the system has no invariant zeros, then $\forall z \in \mathbb{C}$, $\text{rank}(P(z)) = n + p$. According to Lemma 1, the system is SIO.

Suppose the digraph of a structured system satisfies Assumption 1, the digraph can be decomposed into three sub-graphs through a separator-based decomposition [15]. The corresponding matrix pencil $P_i(z)$ is divided into three subsystem pencils

$$P_i(z) = \begin{bmatrix} P_{11}(z) & * & * \\ \mathbf{0} & P_{22}(z) & * \\ \mathbf{0} & \mathbf{0} & P_{33}(z) \end{bmatrix}. \quad (14)$$

where $P_{11}(z)$ generically has full row normal rank, even after the deletion of an arbitrary column; $P_{22}(z)$ generically is invertible; and $P_{33}(z)$ generically has full column normal rank, even after the deletion of an arbitrary row. The $*$'s denote matrices (matrix pencils) that are not relevant in the context of this paper. The number of invariant zeros of the system is the sum number of invariant zeros of three subsystems.

Since $\mathcal{G}_0$ satisfies Assumption 1, The system in (11) is SIO, therefore, there exists no invariant zeros in $P_{22}(z)$ and $P_{33}(z)$. Assume the system contains a subsystem $P_{11}(z)$, which has full row rank. There are more input vertices than output vertices in the subsystem, and $P_{11}(z)$ does not have full column rank. There exists a state-zero direction x_0 , $x_0 \neq 0$, and an input-zero direction g , such that

$$P_{11}(z) \begin{bmatrix} x_0 \\ g \end{bmatrix} = 0. \quad (15)$$

The nontrivial zero dynamics cannot be used to recover the initial state of the system. The subsystem is not SIO. Therefore, the matrix pencil in (11) does not include a full-row rank subsystem pencil. $P_{22}(z)$ is invertible, and the subsystem pencil has full column rank. $P_{33}(z)$ is a matrix with full column rank. The normal rank of matrix pencil is $n + p$.

For the system (11), the graph is connected and every vertex in the graph has a self-loop. A sufficient and necessary condition for the state and input observability is derived.

Theorem 1. Given a subsystem in (11) and the corresponding graph $\mathcal{G}_0$, the system is SIO if and only if all the vertices in direct $\mathcal{I}\mathcal{O}_i - S^*(\bar{\mathcal{I}}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i)$ paths in $\mathcal{G}_0$ belong to $\mathcal{V}_{\text{ess}}(\bar{\mathcal{I}}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i)$.

Proof. We first prove the sufficiency of the theorem.

Sufficiency: since all the vertices in the linking from $\mathcal{I}\mathcal{O}_i$ to $S^*(\bar{\mathcal{I}}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i)$ in $\mathcal{G}_0$ belong to $\mathcal{V}_{\text{ess}}(\bar{\mathcal{I}}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i)$, the vertex in $\bar{\mathcal{I}}\mathcal{O}_i$ are covered by all the maximum size of linking. Therefore, $\rho(\mathcal{I}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i) = |\mathcal{I}\mathcal{O}_i|$, and the matrix pencil $P_i(z)$ has full column rank. Using the minimum separator $S^*(\bar{\mathcal{I}}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i)$, the matrix pencil of the system can be divided as follows

$$P_i(z) = \begin{bmatrix} P_{11}(z) & * \\ \mathbf{0} & P_{22}(z) \end{bmatrix} \quad (16)$$

where $P_{11}(z)$ is invertible, and $P_{22}(z)$ has full column rank even after deleting an arbitrary row. The $*$ denotes matrices that are not relevant in the context.

From the construction of $\mathcal{G}_0$, every state vertex in set $\bar{\mathcal{I}}\mathcal{O}_i$ has a self-loop. If we simply select a maximum linking from $S^*(\bar{\mathcal{I}}\mathcal{O}_i, \bar{\mathcal{I}}\mathcal{O}_i)$ to $\bar{\mathcal{I}}\mathcal{O}_i$, and include all the self-loops that are not contained in the linking, the disjoint set of vertices contain all state vertices in $\bar{\mathcal{I}}\mathcal{O}_i$. According to Lemma 4, there exist no invariant zeros in $P_{22}(z)$.

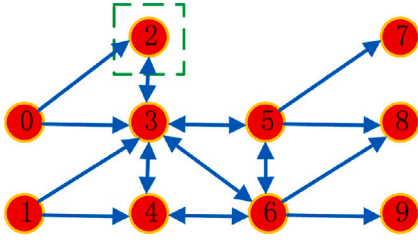


Fig. 2. Digraph representing the system of Example 2.

All the state vertices between $I\bar{\mathcal{O}}_i$ and $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ in $\mathcal{G}_0$ belong to $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. It is clear that all the state vertices in $P_1(z)$ are included in $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. Every maximum linking contains all the state vertices. According to Lemma 3, $P_{11}(z)$ has no invariant zeros. Therefore, $P(z)$ has no invariant zeros, and the structured system is SIO.

Necessity: According to the discussion above, if the system is SIO, the matrix pencil of the system does not consist of a sub-matrix block which has full row rank, and there exist no invariant zeros. Therefore, the normal rank of the system is $|I\bar{\mathcal{O}}_i| + |\bar{I}\mathcal{O}_i|$. From Lemma 2, the maximum size of a linking from $I\bar{\mathcal{O}}_i$ to $\bar{I}\mathcal{O}_i$ equals $|I\bar{\mathcal{O}}_i|$. The graph $\mathcal{G}_0$ can be partitioned into two sub-graphs by $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. The two sub-graphs are disconnected and have no invariant zeros. $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ is a minimum separator and its cardinality is equal to the size of the maximum linking. The sub-matrix block represented by the graph between $I\bar{\mathcal{O}}_i$ and $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ are invertible. According to the number of invariant zeros of an invertible matrix pencil, every vertex is covered by any maximum size of linking from $I\bar{\mathcal{O}}_i$ to $\bar{I}\mathcal{O}_i$. Thus all the vertices in direct $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths of $\mathcal{G}_0$ belong to $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. $\square$

From Theorem 1, it can be concluded that the sub-graph formed by all the vertices in direct $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths determine whether the system is SIO. Therefore, the positions of vertices in set $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ are critical to analysis. If $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ and $I\bar{\mathcal{O}}_i$ are identical, then the corresponding system is SIO. Adding extra output vertices can influence the position of set $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. In the following subsection, a specific vertex addition scheme is introduced to recover the state and input observability.

4.2. Observability recovery by sensor addition

In practical applications, graphs of multi-agent systems are sparsely connected, and the proportion of agents that can be directly measured is small. To improve the performance of the system in a malicious environment, let two adjacent agents in the system share their local observations during state recovery. Two-hop information can be utilized without extra hardware implementation.

Remark 3. Information sharing must be carried out between two well-protected agents. Since agents in the system can be hijacked by attackers, sharing information with arbitrary agents may lead to information leakage. Besides, a hijacked agent may give modified information to a normal agent, which makes it much more difficult to recover the initial states of the system.

Assume agents i and j are two adjacent agents with high-security level. They share information with each other, and the output of the system is given as

$$y[k] = \begin{bmatrix} C_i \\ C_j \end{bmatrix} x[k]. \quad (17)$$

Two agents share their local observations, and the states of agents in $\mathcal{N}_i \cup \mathcal{N}_j$ can be directly measured. The cardinality of the set is $|\mathcal{N}_i \cup \mathcal{N}_j| = |\mathcal{N}_i| + |\mathcal{N}_j| - |\mathcal{N}_i \cap \mathcal{N}_j|$. Therefore, observation sharing between two agents implements more independent information. Considering two extreme situations, agents i and j have the same neighbors

and totally different neighbors. When the neighbors of agents i and j are the same, information sharing makes no difference at all. When two agents have no common neighbors, the cardinality of the measurable states becomes $|\mathcal{N}_i| + |\mathcal{N}_j|$.

Remark 4. If agents i and j form a dominating set of the graph, then the state of every agent can be directly measured. This is equivalent to a global monitor. Regardless of the number of attacked agents and the vertex connectivity, the system is SIO.

Next, the influence of information sharing is investigated. Repeated information cannot improve the performance of the system. Therefore, information sharing can be viewed as implementation of extra agents that can be directly measured in the system. The original observed agents are in set $\mathcal{N}_i$, and the additional observed agents are in set $\mathcal{N}_j \setminus (\mathcal{N}_i \cap \mathcal{N}_j)$. The state-space representation of the multi-agent system which implements information sharing is:

$$\begin{aligned} x[k+1] &= Wx[k] + Bu[k], \\ y[k] &= C_i x[k], \\ z[k] &= C_e x[k]. \end{aligned} \quad (18)$$

where the matrix C_i corresponds to agents in $\mathcal{N}_i$, and C_e corresponds to agents implemented by information sharing.

The effect of information sharing can be viewed as a special output vertex addition procedure. The neighbors of agent i depend on the graph topology. Additional vertices depend on the selection of agent j . For ease of analysis, the situation where arbitrary output vertices can be directly implemented in the system is considered.

When a multi-agent system does not satisfy the condition in Proposition 1, it is not SIO. There are not enough output vertices such that the initial states can be recovered regardless of unknown inputs. Extra output vertices can be added to the system to obtain more independent observation equations to determine the initial states.

4.2.1. A single vertex

If there exists a vertex x_i in all direct $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths, such that $x_i \notin \mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$, then the system is not SIO.

If $x_i \in I\bar{\mathcal{O}}_i$ and $x_i \notin \mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$, an extra output vertex can be directly added to the vertex. The state of x_i can be directly measured and $x_i \in I\bar{\mathcal{O}}_i$, therefore, x_i can be viewed as a vertex in set $I\bar{\mathcal{O}}_i$. The vertex in set $I\bar{\mathcal{O}}_i$ has no influence of the state and input observability.

If $x_i \in \bar{I}\mathcal{O}_i$ and $x_i \notin \mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$, all other vertices within direct $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths belong to $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. Therefore, all vertices in $I\bar{\mathcal{O}}_i$ belong to $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. The maximal size of linking from $I\bar{\mathcal{O}}_i$ to $\bar{I}\mathcal{O}_i$ is $|I\bar{\mathcal{O}}_i|$. The matrix pencil of the system has full column rank and can be divided by the minimum separator $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ as follows

$$P(z) = \begin{bmatrix} P_{11}(z) & * \\ \mathbf{0} & P_{22}(z) \end{bmatrix} \quad (19)$$

where $P_{11}(z)$ is invertible and represents the sub-graph formed by direct $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths. And $P_{22}(z)$ has full column rank even after deleting an arbitrary row and represents the sub-graph formed by $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i) - \bar{I}\mathcal{O}_i$ paths. The sub-matrix block $P_{22}(z)$ has no invariant zeros, and extra output vertex is unnecessary. An extra output vertex can be directly added to x_i , the resulted system is SIO. Adding an extra output vertex leads to an extra row in matrix pencil $P_{11}(z)$. Since $P_{11}(z)$ is invertible, the matrix pencil with an extra row $P'_{11}(z)$ still has full column rank. The deletion of an arbitrary row of $P'_{11}(z)$ leads to a full-column rank matrix, then $P'_{11}(z)$ has no invariant zeros.

Example 2. Considering the system represented by the graph in Fig. 2, vertices 0 and 1 are in $I\bar{\mathcal{O}}_i$, vertices 7, 8, and 9 are in $\bar{I}\mathcal{O}_i$, and the remaining vertices are state vertices. From the graph, the maximal size of vertex-disjoint linking from set $I\bar{\mathcal{O}}_i$ to $\bar{I}\mathcal{O}_i$ is two, which indicates a minimal separator contains two vertices. All minimum separators between $I\bar{\mathcal{O}}_i$ and $\bar{I}\mathcal{O}_i$ form the essential vertices, and $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i) =$

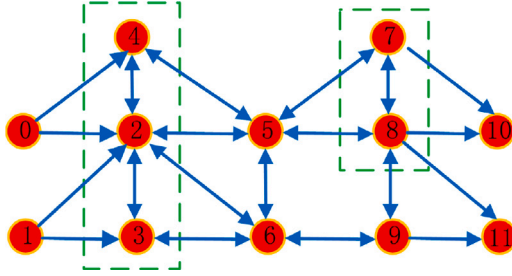


Fig. 3. Digraph representing the system of Example 3.

$\{0, 1, 3, 4, 5, 6\}$. A special minimum separator S^* is defined as the end vertices of all direct $\mathcal{V}_1 - \mathcal{V}_{ess}(\mathcal{V}_1, \mathcal{V}_2)$ paths, and $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i) = \{5, 6\}$. Except for vertex 2, the vertices in $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths belong to $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$. The system is not SIO, and it has one invariant zero. Adding an extra output vertex to vertex 2, it can be verified that the system is SIO and has no invariant zeros.

4.2.2. A vertex set

If there exist a set of vertices $\mathcal{X}_{\mathcal{O}}$ in direct $I\bar{\mathcal{O}}_i - S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ paths, such that $\mathcal{X}_{\mathcal{O}} \not\subset \mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$, the system is not SIO. The state and input observability can be recovered by adding extra output vertices to each vertex in $\mathcal{X}_{\mathcal{O}}$. However, it is unnecessary to implement that many extra vertices.

Assuming $\mathcal{X}_{\mathcal{O}}$ can be divided into multiple strongly connected components $\{\delta_1, \delta_2, \dots, \delta_t\}$, and for vertices in each subset δ_i , they are connected by a direct edge. Adding an extra output vertex to a vertex in δ_i , the remaining vertices in δ_i also have a linking to the output vertex. The state and input observability can be recovered by adding an extra output vertex to any vertex in each subset δ_i .

Given a multi-agent system with known topology, alternative solution which implements less number of additional measurable agents can be found. In this paper, a necessary and sufficient condition of strong observability is proposed, and a minimum separator decomposition can be used to determine whether the modified system which obtains more information satisfy the condition. Therefore, the proposed observability recover procedure can be optimized. A summary table of the sensor addition strategy is provided in Table 1.

Example 3. Considering the system represented by the graph in Fig. 3, vertices 0 and 1 are in $I\bar{\mathcal{O}}_i$, vertices 10 and 11 are in $\bar{I}\mathcal{O}_i$, the remaining vertices are state vertices. From the graph, the maximal size of vertex-disjoint linking from set $I\bar{\mathcal{O}}_i$ to $\bar{I}\mathcal{O}_i$ is two, which indicates a minimal separator contains two vertices. All minimum separators between $I\bar{\mathcal{O}}_i$ and $\bar{I}\mathcal{O}_i$ form the essential vertices, and $\mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i) = \{0, 1, 5, 6, 9, 10, 11\}$. A special minimum separator S^* is defined as the end vertices of all direct $\mathcal{V}_1 - \mathcal{V}_{ess}(\mathcal{V}_1, \mathcal{V}_2)$ paths, and $S^*(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i) = \{10, 11\}$. The vertices 2, 3, 4, 7, and 8 form the set $\mathcal{X}_{\mathcal{O}}$. It is clear that $\delta_1 = \{2, 3, 4\}$, and $\delta_2 = \{7, 8\}$. The system has two invariant zeros, and it is not SIO. Adding two extra output vertices to any vertex of the two subsets, i.e. by implementing extra output vertices to vertices 4 and 7, it can be verified that the condition for the state and input observability is satisfied.

Remark 5. The sensor addition procedure is not actually using measurement from any other additional agent. To solve the strong observability recovering problems, multi-hop information which can be obtained from neighbor agents is utilized. It is worth noting information sharing cannot ensure the recovery of state and input observability of a multi-agent system. Due to the diversity of graph topology, information sharing may not lead to an extra output vertex addition, besides the added vertices may not be placed at the proper location. Nevertheless, it is shown that extra output vertices can turn a system that is not SIO into an observable one. By implementing sufficient output vertices, the state and input observability is guaranteed.

Table 1

A summary table of the sensor addition strategy.

Step	Operation
1	Partition the graph into four mutually disjoint subsets
2	Find special subgraphs by a minimum separator decomposition
3	Test whether the system is SIO
4	A single vertex $x_i \notin \mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$ A vertex set $\mathcal{X}_{\mathcal{O}} \not\subset \mathcal{V}_{ess}(I\bar{\mathcal{O}}_i, \bar{I}\mathcal{O}_i)$
5	Add a sensor to the vertex Divide $\mathcal{X}_{\mathcal{O}}$ into connected components Add sensors to each subset
6	The strong observability is recovered

5. Conclusions

This paper investigated the state and input observability problem of multi-agent systems. A sufficient and necessary condition was obtained from a graph-theoretic perspective, and the derived results have shown that additional output vertices can be utilized to recover the state and input observability. Moreover, we have analyzed the effect of observation sharing between adjacent agents, which can improve the performance of multi-agent systems without additional hardware implementation.

CRedit authorship contribution statement

Xiufeng Zhang: Conceptualization, Writing – original draft. **Jian Sun:** Funding acquisition, Writing – review & editing, Validation, Supervision, Project administration. **Gang Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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